SMART INDIA HACKATHON 2026

TRAJECTORY

AI-Based Network Attack Forecasting from Network Traffic Data

An offline, explainable temporal cyber-defence system that forecasts likely attack progression before compromise is complete.

Network Telemetry → **Current State** → **Future Forecast** → **Earlier Response**

PROBLEM STATEMENT ID
SIH26153

PROBLEM STATEMENT TITLE
AI based Network Attack Forecasting from Network Traffic Data

ORGANIZATION
National Technical Research Organisation (NTRO)

THEME
Blockchain & Cybersecurity

PS CATEGORY
Software

TEAM ID
[ADD OFFICIAL TEAM ID]

TEAM NAME
[ADD OFFICIAL TEAM NAME]

INSTITUTE
[ADD OFFICIAL INSTITUTE NAME]

Forecasts are probabilities with evidence; Trajectory supports analyst decisions and does not block traffic.

Solution Overview & Prototype

From static alerts to predictive attack intelligence

Trajectory ingests flow CSV and PCAP-derived packet data, converts them into ordered network states, and forecasts likely attack progression over K future windows: risk, predicted stage, affected assets, confidence and supporting evidence in an offline analyst dashboard (Web/desktop).

SOC Analyst

- Load CSV flows / PCAP captures
- Risk timeline over time windows
- Next likely stage + affected assets
- Evidence behind every forecast
- Predicted vs actual replay

Incident Responder

- Prioritize hosts/segments at risk
- Confidence + insufficient-evidence flags
- Decision support, not auto-blocking
- Export reproducible forecast record

Security Lead / Evaluator

- Temporal model vs logistic baseline
- Precision, recall, F1, FPR, calibration
- Forecast lead time
- Deterministic, seed-fixed replay
- Fully local; no cloud AI

Project Status

- Phase: ingestion + temporal-data foundation done
- CSV & PCAP ingestion, flow + packet features
- Timestamped overlapping state windows
- Transition targets, leakage-safe splits
- 15 tests passing; baseline model is next

Why We Stand Out

- Predictive, not retrospective
- K-step future-state simulation
- Temporal context instead of per-flow labels
- Evidence + assets with every forecast
- Offline-first; honest baseline comparison

PROTOTYPE VISUALS

1. Data Workspace

CSV/PCAP input | event count, time range | feature-coverage indicators

2. Forecast Dashboard

Risk over K windows | Current / Forecast / Actual replay | stage + confidence

3. Evidence View

Driving features | Server-03, failed-auth burst, destination diversity | source events

ILLUSTRATIVE OUTPUT

```
Current state: Suspicious reconnaissance
Predicted stage: Lateral movement  (+3 windows)
Likely asset: Server-03      Confidence: 0.72
Evidence: failed-auth burst, new internal
connections, rising destination diversity
```

Replace mockup panels with screenshots and the illustrative output with model output before submission.

Backend Architecture & Technical Approach

Local, reproducible, evidence-first forecasting pipeline

INPUT	Flow CSV | PCAP captures -> normalization + packet feature extraction
DATA	Unified events -> timestamped windows -> state store | labels, transition targets, split manifests
MODEL	Logistic Regression baseline (current window) | GRU/LSTM temporal model + K-step rollout
OUTPUT	Risk timeline | attack stage | assets | evidence | confidence + warnings
UI	Offline analyst dashboard (Streamlit/Flask + Plotly)

Security & Privacy

- Local-first; no external AI service
- Anonymized identifiers in demos
- No secrets/captures committed
- Provenance, versions, checksums
- Human-in-the-loop; no auto-blocking

Input Fallback Logic

- PCAP -> flow + packet features
- CSV only -> flow features + packet-coverage warning
- Data missing -> validation error, no fabricated forecast
- All paths run offline

TECHNICAL APPROACH FLOW

Ingest — Normalize — Window — Prepare

Learn — Simulate — Explain — Display

Security lead configures scenario & evaluation -> Analyst loads telemetry & inspects forecast -> Responder validates evidence & acts

AI COMPONENTS

Component	Technique	Function
Static baseline	Logistic Regression	Fair reference from the current window
Temporal forecaster	GRU / LSTM (proposed)	Learn dependencies across ordered states
Future simulation	Recursive K-step rollout	Likely future states, risk trajectory
Stage mapping	Documented rules / classifier	MITRE ATT&CK-oriented stage label
Explanation	Attribution + event retrieval	Driving features and traffic evidence
Calibration	Reliability analysis	Useful confidence values

No Kafka/cloud gateway/microservices are claimed: the prototype runs on local files and model artifacts. Docker only if packaged before submission.

Feasibility, Viability & Challenges

Feasible with open data, local compute and controlled deployment

Technical

- Public flow & PCAP datasets available
- Open-source parsers + ML frameworks
- GRU/LSTM practical on dev hardware
- Scenario-safe splits, baseline defined

Operational

- Runs beside IDS / SIEM / EDR
- Analyst owns response decisions
- Deterministic replay for demos & audit
- Local processing for sensitive sites

Economic

- Open-source stack, low prototype cost
- Software-only MVP, no custom hardware
- Augments existing tools
- Pilot-first phased adoption

Regulatory & Privacy

- Synthetic / public / anonymized demo data
- Licences & access restrictions recorded
- Captures and model artifacts stay local
- Human approval; no autonomous enforcement

Market growth (illustrative shape)

CAGR: [ADD VERIFIED VALUE]% Source: [ADD REPORT, PUBLISHER, YEAR, URL]

Chart shape is illustrative until a verified market source is inserted; state geography and product scope with the figure.

CHALLENGES -> TECHNICAL RESPONSES

Challenge	Technical response
Rare, imbalanced attacks	Class weighting, per-class metrics, PR analysis
Temporal leakage	Scenario / campaign / time-held-out evaluation
Dataset shift	Secondary-dataset tests, per-site recalibration
Missing packet features	Coverage report + reduced-context warning
Recursive forecast drift	Short K horizon, calibration, uncertainty display
Ambiguous stage labels	Documented mapping; insufficient-evidence state
False-positive fatigue	Baseline comparison, threshold tuning, FPR reporting
Black-box predictions	Attribution, source-event evidence, asset context
Sensitive telemetry	Local processing, anonymization, restricted artifacts
Enterprise integration	File-based MVP first; adapters/APIs later

Viability: feasible as an offline prototype on public datasets and open-source tools. Production use needs environment-specific validation, integration, calibration, access control and security review.

Impacts, Benefits & Stakeholder Scenario

Earlier, evidence-backed decisions can reduce attack impact

Economic

- Analyst time on likely paths
- Less manual correlation
- Earlier intervention, lower downtime cost
- No stack replacement

Social

- Resilient public services & CII
- Clearer incident context
- Humans keep the decision
- Accountable AI use

Environmental

- Runs on ordinary hardware
- Reuses existing telemetry
- No mandatory cloud transfer
- Software-only, no devices

Operational

- Forecast before next stage
- Stage + asset + horizon + evidence
- Repeatable replay for audit
- Layer on existing controls

SUSTAINABLE DEVELOPMENT GOALS

SDG 9

Industry, Innovation & Infrastructure

Resilience of digital and critical infrastructure via predictive security analytics

SDG 16

Peace, Justice & Strong Institutions

Safer digital public systems; accountable, evidence-backed security operations

Use official icons only where template/UN guidance permits.

SAMPLE SCENARIO: RECONNAISSANCE -> LATERAL MOVEMENT

Workstation-17 contacts many new destinations → **Trajectory correlates recon + failed-auth burst** → **SOC analyst: early forecast toward Server-03** → **Responder validates evidence, acts** → **Security lead audits forecast vs outcome**

Success criteria: forecast appears before the target stage is observable; evidence names real patterns and entities; predicted vs actual are visually distinct; replay is reproducible from the same configuration.

QUANTITATIVE TARGET IMPACT

```
Forecast lead time = t(target stage observable) - t(warning threshold crossed)
Time saved / incident = mean investigation time (current) - mean time with
evidence panel
Analyst-hours recovered / year = valid incidents per year x time saved per
incident
```

[ADD X] min median forecast lead time

[ADD Y]% investigation-time reduction (controlled user test)

[ADD Z] analyst-hours recovered / year | basis: [ADD DATASET / SCENARIOS / METHOD]

Do not convert model accuracy into financial savings without an explicit operational assumption.

Research, Market Sizing & Business Model

Research-backed; scalable from offline pilot to enterprise integration

Official & Domain

- SIH 2026 PS: sih.gov.in/sih2026PS
- MITRE ATT&CK: attack.mitre.org | CAPEC: capec.mitre.org
- NVD: nvd.nist.gov | NCIIPC: nciipc.gov.in

Datasets

- CIC-IDS2017/2018, UNSW-NB15, CTU-13, CICIoT2023, LANL auth
- URLs: [ADD VERIFIED DATASET URLS]
- Only with documented licence, access, checksums

AI / ML Research

- GRU/LSTM sequence forecasting; recursive multi-step + uncertainty
- Calibration; feature attribution / XAI; scenario-held-out IDS evaluation
- Papers: [ADD 2-4 AUTHOR, TITLE, VENUE, YEAR, DOI]

Software

- PyTorch | scikit-learn | Scapy | NetworkX | Plotly | Streamlit

MARKET SIZING (BOTTOM-UP)

Tier	Definition	Estimate
TAM	Orgs with network-security teams in target geography x annual value	[ADD N1 x A = TAM]
SAM	Reachable with offline product + supported integrations	[ADD N2 x A = SAM]
SOM	Realistic customers in first 3 years	[ADD N3 x A = SOM]

State: geography & sectors, org-count source, scope of annual value, adoption rationale, tax/support inclusion.

REVENUE & HARDWARE

Component	Unit	Pricing basis
Platform licence	Per org / env / year	[ADD VALIDATED PRICE]
Deployment & integration	One-time per env	Data sources + SIEM effort
Model calibration	Per env / telemetry change	Data prep + threshold tuning
Support & updates	Annual	% of licence or tier
Hardware	Not required for MVP	Customer workstation/server

FIRST-YEAR REVENUE (ESTIMATE)

```
Licences   [ADD L] x Rs[licence]   = Rs[R1]
Deploys    [ADD D] x Rs[deploy]    = Rs[R2]
Support    [ADD S] x Rs[support]   = Rs[R3]
Training   [ADD T] x Rs[training]  = Rs[R4]
------------------------------------
Total (business estimate) = Rs[ADD TOTAL]
```

PROOF DOCUMENTS

- Source repo: [ADD GITHUB URL]
- Architecture: ARCHITECTURE.md
- Status: IMPLEMENTATION_STATUS.md
- Evaluation report: [ADD LINK AFTER BENCHMARKS]
- Demo video: [ADD FINAL VIDEO LINK]
- Consolidated research: [ADD FOLDER LINK]
- Reproducibility: README.md + uv.lock

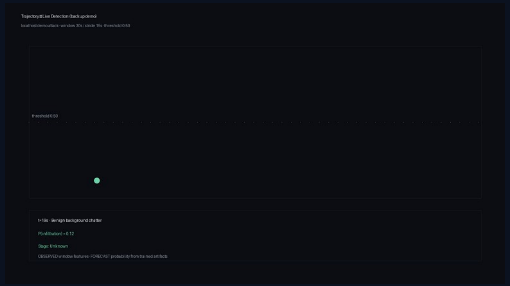
Trajectory adds a predictive layer to existing cyber defence: what may happen next, where, and why the analyst should look.

Live Demo — Watch the AI Detect an Attack

Real-time detection on a localhost attack simulation; deterministic and rehearsed

Stage Storyline (measured in rehearsal)

- 0:00 — Train the models live (~3 seconds, deterministic seed)
- 0:10 — Press Start: benign chatter begins
- 0:19 — Quiet window: P(infiltration) = 0.12, no false alarms
- 0:34 — ALERT: scan burst detected at P = 0.99
- Stage: Initial Access — MITRE ATT&CK TA0001
- 0:49 — Failed logins: P = 1.00, evidence panel grows
- 1:04 — ESCALATION: Lateral Movement — MITRE TA0008
- Same trained artifacts as the offline benchmark
- Sources: demo attack, CIC-IDS2017 replay, live capture



t=19s — benign, P=0.12

t=34s — ALERT, P=0.99

t=64s — Lateral Movement

Backup if live infra fails: deliverables/backup_demo/backup_demo.gif (animated) — same numbers as the live run.